Transportation problem resolution (continuation)

2 and **3** - Two methods can be used:

- **Stepping-Stone method**
  (the logic is the same of the Simplex method)

- **Dantzig** method (or **transportation method**)
  (it is based on duality)

The Dantzig method is more efficient from the computation point of view (it involves less operations per iteration), therefore this is the method that is following described.

Example

Let’s consider the previous example (page VI-5) with the initial **FBS** obtained by the **Northwest Corner method**.

	D1	D2	D3	D4	
O1	12 4	3 3	8	4	7
O2	7	4 5	6 5	9	10
O3	8	7	3 6	6 6	12
	4	8	11	6	

1. Starting from the primal problem we derive the dual problem:

Primal

$$\text{Minimize } z = 12\,x_{11} + 3\,x_{12} + 8\,x_{13} + 4\,x_{14} + 7\,x_{21} + 4\,x_{22} + 6\,x_{23} + 9\,x_{24} + 8\,x_{31} + 7\,x_{32} + 3\,x_{33} + 6\,x_{34}$$

subject to

$$\begin{array}{llllllllllllll}
x_{11}+x_{12}+x_{13}+x_{14} & & & = & 7 \\
 & x_{21}+x_{22}+x_{23}+x_{24} & & = & 10 \\
 & & x_{31}+x_{32}+x_{33}+x_{34} & = & 12 \\
x_{11} & +x_{21} & +x_{31} & = & 4 \\
x_{12} & +x_{22} & +x_{32} & = & 8 \\
x_{13} & +x_{23} & +x_{33} & = & 11 \\
x_{14} & +x_{24} & +x_{34} & = & 6
\end{array}$$

$x_{ij} \geq 0$; $i = 1, 2, 3$; $j = 1, 2, 3, 4$

Corresponding Dual

Maximize $z = 7 u_1 + 10 u_2 + 12 u_3$
$+ 4 v_1 + 8 v_2 + 11 v_3 + 6 v_4$

subject to

$u_i + v_j \leq c_{ij}$ (12 constraints)

u_1, u_2, u_3 without sign constraint

(associated to supply constraints)

v_1, v_2, v_3, v_4 without sign constraint

(associated to demand constraints)

The simplicity of the previous model can be observed, where each constraint contains only two variables related to the respective origin and destination.

2. The constraints of the dual corresponding to the occupied cells of the table (O1D1, O1D2, O2D2, O2D3, O3D3, O3D4) are written but in the equality form:

$$\begin{cases} u_1 + v_1 = 12 \\ u_1 + v_2 = 3 \\ u_2 + v_2 = 4 \\ u_2 + v_3 = 6 \\ u_3 + v_3 = 3 \\ u_3 + v_4 = 6 \end{cases}$$

Dantzig system

- Six equations with seven unknown variables
- We assign a zero value to any of the variables, usually u_1, and we solve the system in order to the others. Thus, with $u_1=0$, the resolution of the system results in:

$$\begin{cases} v_1 = 12 \\ v_2 = 3 \\ u_2 = 1 \\ v_3 = 5 \\ u_3 = -2 \\ v_4 = 8 \end{cases}$$

Alternatively, this solution can be obtained directly from the transportation table. For this, a new column is created for the variables ui and a new row for the variables vj (see the following table). Then, by setting u1=0, one must move in line through the cells corresponding to the basic variables, to obtain vj, and, given these values, one must move in column through the cells corresponding to the basic variables, to get the ui.

	$v_1 = 12$	$v_2 = 3$	$v_3 = 5$	$v_4 = 8$	
	D1	D2	D3	D4	
O1 $u_1 = 0$	12 4	3 3	8	4	7
O2 $u_2 = 1$	7	4 5	6 5	9	10
O3 $u_3 = -2$	8	7	3 6	6 6	12
	4	8	11	6	

3. We test, with these values, the constraints corresponding to each of the empty cells of the current solution (O1D3, O1D4, O2D1, O2D4, O3D1, O3D2):

$$\begin{cases} u1 + v3 \leq 8 \Leftrightarrow 5 \leq 8 \text{ (V)} \checkmark \\ u1 + v4 \leq 4 \Leftrightarrow 8 \leq 4 \text{ (F)} \times \\ u2 + v1 \leq 7 \Leftrightarrow 13 \leq 7 \text{ (F)} \times \\ u2 + v4 \leq 9 \Leftrightarrow 9 \leq 9 \text{ (V)} \checkmark \\ u3 + v1 \leq 8 \Leftrightarrow 10 \leq 8 \text{ (F)} \times \\ u3 + v2 \leq 7 \Leftrightarrow 1 \leq 7 \text{ (V)} \checkmark \end{cases}$$

The tested solution is not optimal because there are constraints that are not verified.

4. To find a new FBS, we start by studying the several empty cells which constraints are not verified. The cell to fill ($\Leftrightarrow$ the variable that will become basic) is the one that has the highest value $\mathbf{u_i+v_j-c_{ij}}$.

In this example:

- Cell O1D4: $u1 + v4 – c14 = 0 + 8 – 4 = \mathbf{4}$
- Cell O2D1: $u2 + v1 – c21 = 1 + 12 – 7 = \mathbf{6}$
- Cell O3D1: $u3 + v1 – c31 = – 2 + 12 – 8 = \mathbf{2}$

The cell O2D1 is the one that has the highest value $\mathbf{u_i+v_j-c_{ij}}$ $\Rightarrow$ we will move product to this cell $\Leftrightarrow$ the variable that will become basic is x_{21}.

To determine the variable that will leave the basis we must build a **cycle** composed by the variable x_{21} and by basic variables of the current solution, that is, by cell O2D1 and by occupied cells of the current table.

A **cycle** is a sequence of cells of the transportation table, where: (i) each consecutive pair of cells belongs to the same row or column; (ii) there are no more than two consecutive cells in the same row or column; (iii) the first and last cells of the sequence belong to the same row or column; (iv) no cell appears more than once, in the sequence.

	$v_1 = 12$ D1	$v_2 = 3$ D2	$v_3 = 5$ D3	$v_4 = 8$ D4	
O1 $u_1 = 0$	12 4	3 3	8	4	7
O2 $u_2 = 1$	7 *	4 5	6 5	9	10
O3 $u_3 = -2$	8	7	3 6	6 6	12
	4	8	11	6	

If you consider that the product movement is done only between cells on the same row, the amount to be moved to the cell marked with '*' is the **minimum of the maximums** that can be moved in each row of the cycle. In this case it will be: Min {4, 5} = 4

Thus, we can conclude that the variable that will become non basic will be x_{11} (since the corresponding cell will become empty $\Rightarrow x_{11}=0$**)**.

5. Build the new table and repeat all the steps from **2.**, until reach the optimal solution.

	$v_1 = 6$ D1	$v_2 = 3$ D2	$v_3 = 5$ D3	$v_4 = 8$ D4	
O1 $u_1 = 0$	12	3 7	8	4 *	7
O2 $u_2 = 1$	7 4	4 1	6 5	9	10
O3 $u_3 = -2$	8	7	3 6	6 6	12
	4	8	11	6	

Cost = 3*7 + 7*4 + 4*1 + 6*5 + 3*6 + 6*6 = 137

Test of the calculated values of ui e vj, in the <u>empty cells</u>:

$$\begin{cases} u_1 + v_1 \le 12 \Leftrightarrow 6 \le 12 \text{ (V)} ✓ \\ u_1 + v_3 \le 8 \Leftrightarrow 5 \le 8 \text{ (V)} ✓ \\ u_1 + v_4 \le 4 \Leftrightarrow 8 \le 4 \text{ (F)} ✗ \\ u_2 + v_4 \le 9 \Leftrightarrow 9 \le 9 \text{ (V)} ✓ \\ u_3 + v_1 \le 8 \Leftrightarrow 4 \le 8 \text{ (V)} ✓ \\ u_3 + v_2 \le 7 \Leftrightarrow 1 \le 7 \text{ (V)} ✓ \end{cases}$$

The solution is not optimal since the constraint correponding to O_1D_4 is not satisfied. In the next table we will move product to this cell (x_{14} will become basic).

How much to move? Min{7,5,6} = 5. The variable that will become basic non basic x_{23}.

	$v_1 = 6$ D1	$v_2 = 3$ D2	$v_3 = 1$ D3	$v_4 = 4$ D4	
O1 $u_1 = 0$	12	3 / 2	8	4 / 5	7
O2 $u_2 = 1$	7 / 4	4 / 6	6	9	10
O3 $u_3 = 2$	8	7	3 / 11	6 / 1	12
	4	8	11	6	

Cost = 3*2 + 4*5 + 7*4 + 4*6 + 3*11 + 6*1 = 117

Test of the calculated values of ui e vj, in the <u>empty cells</u>:

$$\begin{cases} u_1 + v_1 \le 12 \Leftrightarrow 6 \le 12 \text{ (V)} \checkmark \\ u_1 + v_3 \le 8 \Leftrightarrow 1 \le 8 \text{ (V)} \checkmark \\ u_2 + v_3 \le 6 \Leftrightarrow 2 \le 6 \text{ (V)} \checkmark \\ u_2 + v_4 \le 9 \Leftrightarrow 2 \le 9 \text{ (V)} \checkmark \\ u_3 + v_1 \le 8 \Leftrightarrow 8 \le 8 \text{ (V)} \checkmark \\ u_3 + v_2 \le 7 \Leftrightarrow 5 \le 7 \text{ (V)} \checkmark \end{cases}$$

Since all the constraints are satisfied the previous table is optimal.

<u>Optimal solution</u>:

x12*=2; x14*=5; x21*=4; x22*=6; X33*=11; X34*=1;

remaining x_{ij}=0.

<u>Minimum transportation cost</u>:

z* = 117 MU

However, since the constraint corresponding to O3D1 is satisfied in equality form, there is an **alternative optimal solution**!

As an exercise, find the alternative optimal solution (moving product to **O3D1**) and calculate the corresponding cost.